MD. MASHRUP HASAN

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Portfolio | LinkedIn | GitHub | ResearchGate | Google Scholar

SUMMARY

Geospatial researcher working at the intersection of GIScience, spatial data science, and environmental health, with research spanning climate-sensitive health risks, flood and disaster risk, women's reproductive health in climate-vulnerable settings, climate vulnerability and migration, and land-use dynamics. My research integrates remote sensing, GIS-based spatial modeling, household surveys, and machine learning to examine how environmental exposures and social vulnerability shape health outcomes and inequities across urban, coastal, and rural communities. Through both academic and community-based applied research, I develop exposure assessment frameworks, composite health risk indices, and spatial decision-support tools. My broader research interest lies in leveraging multi-source geospatial data and artificial intelligence to understand how climate change alters environmental health risks, quantify their drivers and disparities, and generate evidence that supports health equity, climate resilience, and disaster risk reduction.

EDUCATION

- **M.S. in Geography and Environment (CGPA 3.86/4.00 with Distinction)** *2023 – 2026*
  Merit Position: 2nd out of 18 students
  - Department of Geography and Environment, Shahjalal University of Science & Technology (SUST), Sylhet, Bangladesh.
  - **Thesis:** *Flood-Induced Health Risk Assessment in Sunamganj: A Deep Learning-Enhanced GIScience Approach.*
  - **Relevant coursework:** Natural Hazards and Disaster Management; Advanced GIS and Remote Sensing; Climate Change, Government Policy and Action Strategy; Water Resource Management; Urban Environmental Management.
- **B.Sc. in Geography and Environment (CGPA 3.70/4.00)** *2019 – 2024*
  Merit Position: 2nd out of 45 students
  - Department of Geography and Environment, Shahjalal University of Science & Technology (SUST), Sylhet, Bangladesh.
  - **Research Project:** *Suitability Analysis for Flood-Adaptive Green Infrastructure: Building Urban Resilience Through Strategic Placement in the Chattogram City Corporation.*
  - **Relevant coursework:** GIS and Remote Sensing; Hydrology and Fluvial Morphology; Climatology; Environmental and Social Impact Assessment; Quantitative Techniques in Geography; Research Methods in Geography and Environment.

RESEARCH AND WORK EXPERIENCE

- **Job:** Center for Participatory Research and Development (CPRD) *Oct 2025 – Present*
  Research Officer *Dhaka, Bangladesh*
  - Conduct mixed-methods research (household surveys, FGDs, and KIIs integrated with climate and health data) on climate-induced health risks as non-economic loss and damage (NELD) among climate migrants in Dhaka's informal settlements and Indigenous Peoples in the Northwest Barind Tract (2026–2027), informing policy analysis of Bangladesh's NAP, H-NAP, NDC, and National Mechanism on Loss and Damage.
  - Completed donor-funded projects on climate justice advocacy for LDCs; climate migration and protection gaps; the layered conflict among climate migrants, Rohingya refugees, and host communities; and climate-induced soil salinity and women's reproductive health.
  - Produce spatial analyses, thematic maps, and 5+ policy reports and briefings for government bodies and international donors, and contribute evidence to CSO position papers and multi-stakeholder consultations for national and UNFCCC advocacy.

- **Job:** Betterskill.io (e-Learning Platform), Bangladesh *Feb 2026 – Present*
  Course Developer & Instructor *Bangladesh*
  - Design and produce a 10-hour recorded course on Kobo Toolbox and NVivo, equipping learners with end-to-end survey design, field data collection, and qualitative analysis workflows.
  - Develop comprehensive course materials including video lectures, slides, datasets, quizzes, and assignments aligned with industry and research standards to ensure practical learning outcomes.
  - Deliver responsive one-to-one learner support by troubleshooting software queries, clarifying methodological concepts, and resolving workflow challenges, ensuring every learner receives targeted guidance tailored to their individual research or professional context.
- **Job:** Nogori, Bangladesh *May 2025 – Present*
  Graphics & App Interface Lead *Bangladesh*
  - Design UI/UX for a live bus tracking service and the StockTrek investment app, taking both from user flows and wireframes to developer-ready mobile screens optimized for quick reading.
  - Design the Turfy and Nogori Master websites, covering page structure, navigation, visual style, and responsive desktop and mobile layouts.
  - Build the Nogori brand portfolio covering logo usage, colour palette, typography, and presentation assets, and produce product, marketing, and social media graphics across the brand system.
- **Job:** Last Line Web Ltd., Sylhet *July 2025 – Sep 2025*
  Product Trainer and Research Assistant *Sylhet, Bangladesh*
  - Designed and delivered hands-on training programs on data collection tools and survey methodologies, raising client proficiency scores by 40% and closing workflow knowledge gaps across user cohorts.
  - Built and managed systematic data collection and MEAL workflows across project cycles, achieving zero overdue submissions over the tenure.
  - Conducted targeted needs assessments and gap analyses, designing structured evaluation criteria to measure training effectiveness, contributing to a 41% improvement in client retention outcomes.
  - Developed standardized report and manual templates that cut production turnaround time by 23% while maintaining version control and professional consistency.
- **Project:** SUST Research Center *Nov 2023 – Jul 2025*
  Research Assistant *Sylhet, Bangladesh*
  - Conducted GIS-based flood modelling, LULC classification, and spatial risk analysis across 3 studies, interpreting satellite imagery and remote sensing composites to quantify human-environmental interactions and urban vulnerability, producing decision-relevant outputs that contributed directly to 4 peer-reviewed journal publications.
  - Collected and validated primary environmental and spatial datasets through fieldwork and remote sensing interpretation, directly strengthening the empirical accuracy of research findings and map-based outputs.
  - Improved the analytical rigour and presentation of 10+ geospatial research manuscripts, ensuring spatial methodologies, figures, and results met institutional and journal submission standards.
- **Job:** College Education Development Project (CEDP), Murari Chand College *Jun 2024*
  Trainer *Sylhet, Bangladesh*
  - Delivered six structured sessions on document processing, graphic design, and video editing to college teachers, building practical digital skills under the Ministry of Education's Basic ICT Training Program.
  - Developed training materials that supported participant learning and ensured sessions were accessible, well-paced, and outcome focused.

TECHNICAL SKILLS

- **GIS & Remote Sensing:** ArcGIS Pro, ArcGIS Desktop, Google Earth Engine.
- **Programming & Scripting:** Python (GeoPandas, Rasterio, Shapely, Scikit-learn), R, HTML, JavaScript, JupyterLab, GitHub.

- **UAV & Photogrammetry:** UAV flight operations, Aerial data acquisition, Image Preprocessing, Ortho mosaic generation, DSM and DEM generation, and UAV-based spatial analysis.
- **Machine Learning & Deep Learning:** SVM, Random Forest, CNN, ANN, YOLO.
- **Data & Survey Tools:** SPSS, NVivo, Kobo Toolbox.
- **Document & Report Production:** Microsoft 365, Google Workspace, LaTeX, Zotero,
- **Design & Multimedia:** Adobe Illustrator, Adobe Photoshop, WordPress, Adobe After Effects.
- **Audio Engineering:** Audacity, FL Studio.

JOURNAL PUBLICATIONS

1. **Hasan, M. M.**, Jaman, F., & Nayem, J. (2026). *Mapping housing vulnerability to hailstorms in Bangladesh: A decision-support approach for disaster risk reduction.* Progress in Disaster Science. https://doi.org/10.1016/j.pdisas.2026.100643
2. Mostan, N., Kabir, M. N., **Hasan, M. M.**, & Islam, M. M. (2026). *Understanding morphological and hydrological changes in the Dhaleshwari River Basin through remote sensing and GIS-based analysis.* International Journal of River Basin Management. https://doi.org/10.1080/15715124.2025.2610998
3. Islam, M., Jaman, F., **Hasan, M. M.**, & Laskor, A. H. (2025). *Impact of extreme rainfall and land use-land cover change on flood hazards in Sylhet District of Bangladesh.* Case Studies in the Environment. https://doi.org/10.1525/cse.2025.2711193
4. **Hasan, M. M.**, Nishat, N. A., Ahsan, S. M. R., & Nayem, J. (2025). *Sylhet walkways as urban windows: Unveiling their role in urban life and environment stewardship.* International Journal of Science and Development, 2(1), 88–96.https://doi.org/10.5281/zenodo.15555477

WORKING PROJECTS

1. Rahman, S., Hossain, M. S., & **Hasan, M. M.** *Spatiotemporal Prediction of Fish Catch in the Bay of Bengal Using Random Forest Regression and Multi-Source Oceanographic Predictors* (Manuscript in preparation)
2. Niam, S., **Hasan, M. M.**, & Uddin, M. M. *Vision-only re-identification of individual plants across repeated UAV passes.* (In revision, Computers and Electronics in Agriculture)
3. Shamsuddoha, M., Anowar, S. B., Rahman, S., Hossain, M. S., Islam, H. R., & **Hasan, M. M.** *Environmental salinity exposure and pelvic inflammatory disease risk among women in climate-vulnerable coastal Bangladesh.* (Manuscript in preparation)
4. Paul, S., **Hasan, M. M.**, Basak, R., & Islam, M. A. *Impact of urbanization on microclimatic suitability in the context of public health: A GIS and RS-based approach in Sylhet City.* (Manuscript in preparation)
5. Koley, N. J., Nayeem, J., Jain, F., **Hasan, M. M.**, & Jaman, F. *Loss and damage impact on climate-induced migration nexus: Experience from climate-vulnerable communities in north-eastern Bangladesh.* (Manuscript in preparation)

CONFERENCE AND SYMPOSIUM

1. **Hasan, M. M.**, Islam, H. M. R., & Al Shariar, M. (2025, June). *Unregulated gher expansion and agrarian transition in Shyamnagar Upazila, Bangladesh: Land use conflict prediction and evidence-based coastal zoning.* International Symposium on Coast, Climate and Community (ISCCC). [Poster Presentation]
2. Talukder, A. A., **Hasan, M. M.**, Nishat, N. A., & Basak, R. (2025, December). *Industrial land conversion and local economic growth: Causal evidence from Olipur, Habiganj.* 1st International Conference on Life and Earth Sciences for Sustainable Development (ICLESSD-2025). [Oral Presentation]
3. Hossain, M. A., Rahman, M. M., Huda, R. B., Sani, M. S. S., & **Hasan, M. M.** (2025, November). *Exploring vertical growth patterns and regulatory compliance of high-rise mixed-use buildings in Dhaka South City Corporation.* International Conference on Geosciences for Achieving Sustainable Development Goals (ICGSDG). [Oral Presentation]
4. Sharmin, I., Nishat, N. A., Jaman, F., Sultan, F., & **Hasan, M. M.** (2025, April). *Carbon storage and vegetation dynamics in Ratargul Swamp Forest: A GIS-remote sensing approach.* 8th International Conference on Engineering Research, Innovation & Education (ICERIE 2025). [Oral Presentation]

5. Talukder, A. A., **Hasan, M. M.**, Sultana, F., Roy, B., & Nishat, N. A. (2025, February). *Spatial dynamics of environment, socioeconomics, and public health in urban transport hubs: A case study in Sylhet City Corporation.* RURS 1st International Student Conference and Research Fair. [Oral Presentation]
6. **Hasan, M. M.**, & Ahmed, T. S. (2022, October). *Sustainable tourism in Bangladesh: Mapping strategies for SDGs.* ExCEEd 2022, Civil and Environmental Engineering Department, SUST. [Poster Presentation]

TECHNICAL PROJECTS

- **RF-CPUE Bay of Bengal** - Random Forest pipeline predicting fisheries CPUE from oceanographic data. *github.com/MashrupHasan/RF-CPUE-BayOfBengal*
  - Built a 5-predictor Random Forest regression pipeline (mixed-layer thickness, salinity, chlorophyll, SST, surface currents) to predict yearly Catch Per Unit Effort across the Bay of Bengal from GLORYS and MODIS raster data, with grouped cross-validation and a Kriged-CPUE variant for full spatial interpolation.
- **CabbageTrack** - Vision-only UAV plant re-identification pipeline. *github.com/MashrupHasan/CabbageTrack*
  - Developed a pipeline combining YOLOv9c-seg detection, phase-correlation visual odometry, and cross-pass image registration to re-identify individual plants across repeated drone passes without GNSS, re-identifying 98.9% of blind-sampled plants (90/91) on a held-out flight.
- **StockTrek** - Multi-account investment platform for DSE/CSE investors. *github.com/MashrupHasan/Investment_Tracker*
  - Built a portfolio platform for Bangladesh stock-market investors that consolidates multiple Beneficiary Owner (BO) accounts, with live DSE price sync, an allocation planner, risk and dividend analytics, and a phone-friendly layout.

EXTRA-CURRICULLAR ACTIVITIES

- **Vice President, GEE Society, Sylhet (Dec 2024 – Jan 2026)**
  Facebook: GEE Society

  Oversaw planning and execution of seminars, training sessions, and awareness programs on Geography and Environment; fostered collaborative learning through mentorship.
- **General Secretary, SHIKORH Cultural Organization, Sylhet (Feb 2023 – Aug 2024)**
  Facebook: SHIKORH

  Led operations for 120+ members; spearheaded large-scale cultural events; coordinated disaster-response charity initiatives.

AFFILIATIONS

- Member, Geography and Environment Department Alumni Network, SUST.
- General Member, SHIKORH, SUST.

PROFESSIONAL TRAININGS

- Environmental Management Systems Awareness (ISO 14001:2015),
  AJA Bangladesh Ltd (Certificate No. AJA-EMS-0525-005) (May 2025)
- Introduction to Programming with Python,
  Institute of Information and Communication Technology (IICT), SUST - under EDGE Project, Bangladesh Computer Council, ICT Division (May–Dec 2024)
- Application of ArcGIS Pro for Regional Planning,
  GEE Society, Shahjalal University of Science & Technology, Sylhet (Nov 2023)
- Research Techniques of Social & Applied Sciences,
  GEE Society, Shahjalal University of Science & Technology, Sylhet (Sep 2023)

- Fecal Sludge Management in Cities: An Element of City-Wide Inclusive Sanitation,
  Civil and Environmental Engineering Department, SUST (Oct 2022)
- Participatory Mapping with OpenStreetMap for Humanitarian and Disaster Preparedness,
  YouthMappers at SUST (Aug 2022)
- Coastal Disasters: Bangladesh Perspective,
  Disaster Research Training and Management Center, University of Dhaka (Jul 2022)

LANGUAGES

Bengali: Native language.
English: Professional Working Proficiency - IELTS Academic Band 7.0 (CEFR C1, Jul 2025).

AWARDS AND HONORS

- Best Oral Presentation Award [3rd], 1st International Conference on Life and Earth Sciences for Sustainable Development (ICLESSD-2025), Jagannath University, Dhaka, Bangladesh. Dec 2025
- National Science and Technology (NST) Fellowship, Ministry of Science and Technology (MoST), Bangladesh. 2025
- 2nd Runner Up, Painting Contest - National Environment Carnival 2023, organized by Youthpreneur Network. 2023
- University Merit Scholarship, Shahjalal University of Science and Technology, Bangladesh. 2019, 2020, 2021

REFERENCES

Dr. Md. Anowarul Islam
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Relationship: Project Supervisor (B.Sc.)

Md Shamsuddoha
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Relationship: Employer and Direct Supervisor